

UV-Induced Extracellular Electrical Responses of Mycelium

Exploratory Analysis of Static Loading Effects

September 18, 2026

Abstract

This report documents a reproducible exploratory analysis of extracellular voltage recordings from mycelium samples subjected to repeated UV illumination, with and without an additional static load. Three mycelium recordings were analysed: one without static loading and two independent recordings with the same nominal static load. An agar control recording was also examined. The analysis follows the acquisition and filtering procedure used in the student's original code, but introduces cycle-aligned, baseline-subtracted response measures and model-independent kinetic descriptors. The results are treated as pilot observations because repeated UV cycles within one recording are not independent biological replicates.

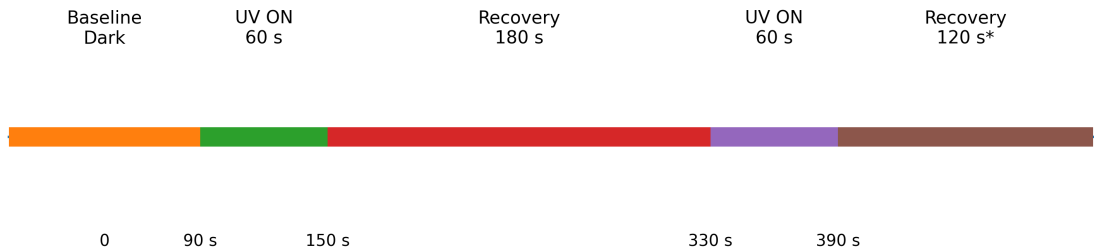
1 Experimental data and protocol

The available datasets are:

- `mycelium_ch1.npy`: no static weight;
- `mycelium_ch1_weight.npy`: static weight, replicate 1;
- `mycelium_ch1_weight2.npy`: the same nominal static weight, replicate 2;
- `agar_control_ch3.npy`: agar control.

The acquisition sampling rate is 100 Hz. The first UV exposure starts at 90 s, with 60 s of UV illumination followed by a nominal 180 s recovery period. This sequence is repeated for 30 UV exposures.

Experimental stimulation protocol



*schematic; final cycle has incomplete recovery in the recordings

Figure 1: Schematic of the UV stimulation protocol used for cycle alignment.

2 Signal processing

The stored voltage is converted from volts to millivolts. For consistency with the student’s original analysis, a Savitzky–Golay filter with a 101-sample window and third-order polynomial is applied. At 100 Hz, the filtering window corresponds to approximately 1.01 s.

Each UV cycle is aligned at UV activation. The preceding 30 s is used as a local baseline, and its median value is subtracted from the complete cycle. This removes experiment-specific DC offsets and makes the comparison focus on the UV-induced change.

For the 29 cycles for which the complete 180 s recovery interval is available, the following quantities are calculated:

$$A_{\max} = \max_{0 \leq t < 60 \text{ s}} \Delta V(t), \quad (1)$$

$$t_{50} : \text{time after the early minimum to reach 50\% of the rise} \quad (2)$$

$$t_{1/2, \text{rec}} : \text{time after UV OFF to reach half the voltage measured immediately before UV OFF}, \quad (3)$$

$$\text{AUC}_{UV} = \int_0^{60 \text{ s}} \Delta V(t) dt. \quad (4)$$

The refined t_{50} definition is intentionally model-independent. For biphasic responses, the early minimum is first identified in the first 20 s; the target is then defined as halfway between this minimum and the median voltage during the final 10 s of UV exposure.

The final UV cycle is omitted from analyses requiring a full recovery interval because the recording does not contain the complete recovery period.

3 Cycle-aligned response

Figure 2 shows the median cycle-aligned response for the three mycelium conditions together with the agar control. The shaded regions around the mycelium curves represent the interquartile range across UV cycles.

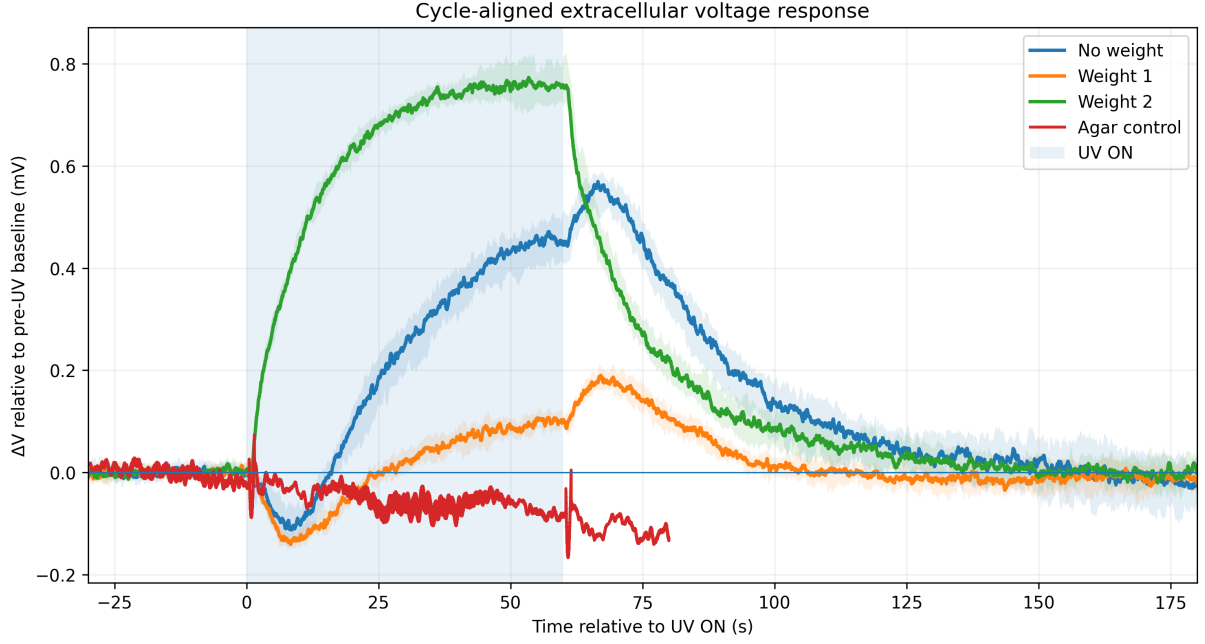


Figure 2: Cycle-aligned extracellular voltage response. Each mycelium curve is the median across available UV cycles; the shaded region is the 25th–75th percentile range. The agar control is shown as a reference waveform.

The recordings exhibit qualitatively different temporal response shapes. The unloaded and first loaded recordings show an initial negative excursion followed by a slower positive response. The second loaded recording shows a predominantly positive response with a substantially faster rise.

The agar control does not reproduce the characteristic sustained positive response observed in the second loaded mycelium recording. Because the agar recording is much shorter than the mycelium recordings, it is treated as a control/reference waveform rather than as a signal to be subtracted from the mycelium measurements.

4 Cycle-to-cycle response

The response amplitude, refined response time, recovery half-time, and integrated UV response are shown in Figures 3–6.

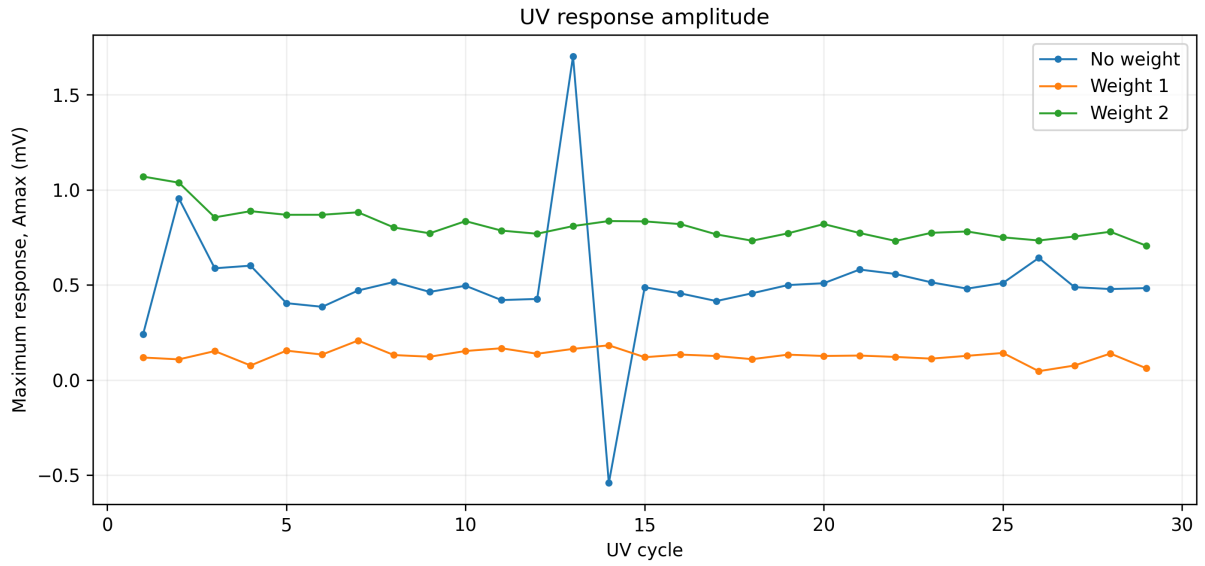


Figure 3: Maximum UV response amplitude for individual UV cycles.

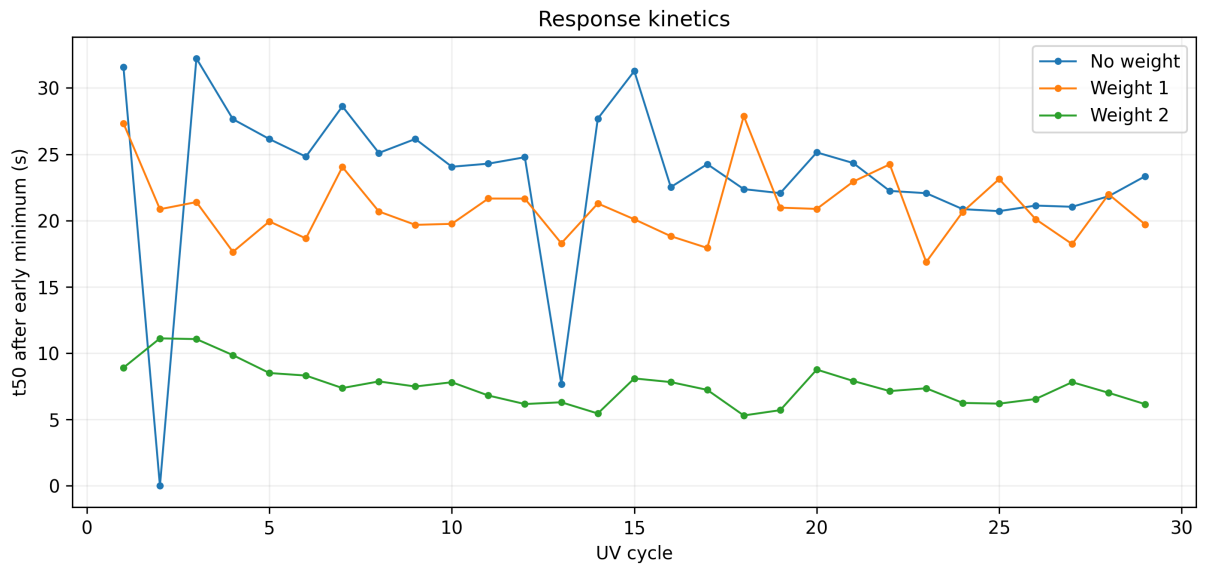


Figure 4: Refined t_{50} for individual UV cycles. The definition is based on the early minimum and the sustained UV response, avoiding a forced single-exponential model for biphasic responses.

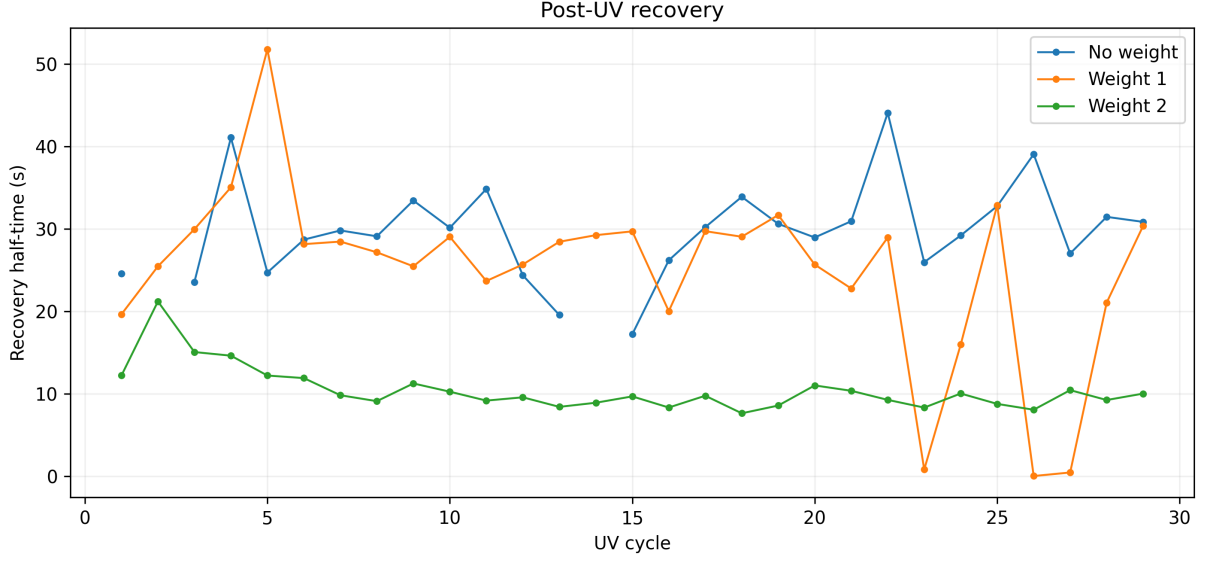


Figure 5: Post-UV recovery half-time for individual cycles.

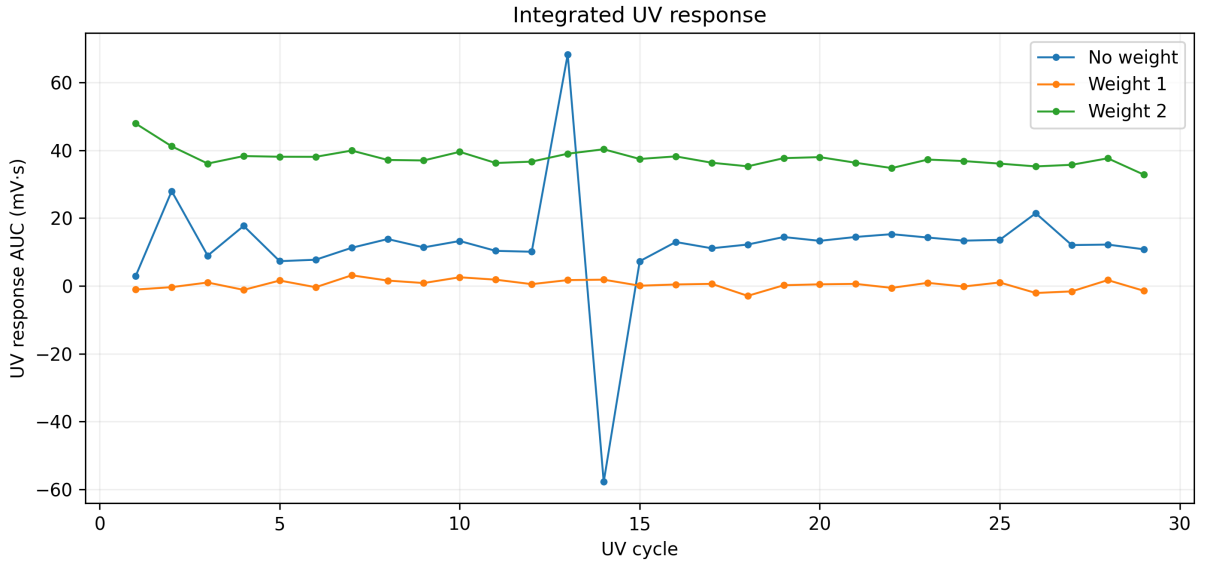


Figure 6: Integrated response during the 60 s UV exposure.

5 Cycle-by-cycle response maps

The heatmaps provide a direct view of the temporal reproducibility of the response. Each row corresponds to one UV exposure and each column to a time relative to UV activation.

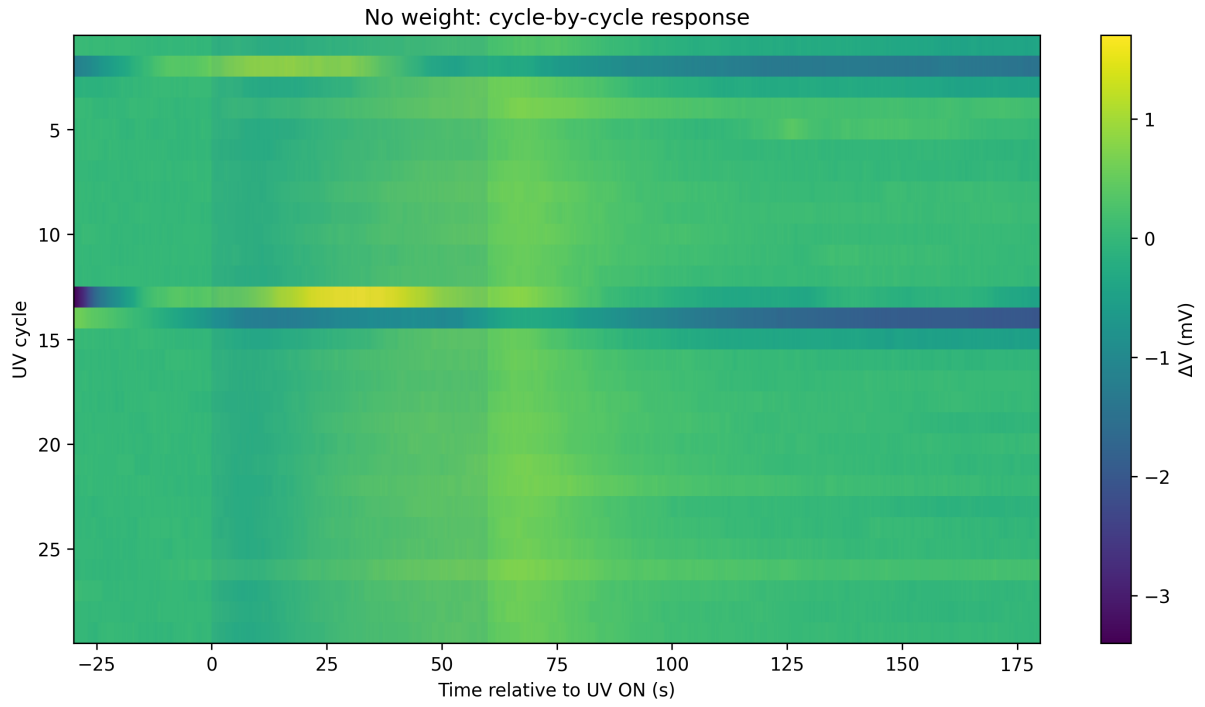


Figure 7: Cycle-by-cycle response matrix for the unloaded recording.

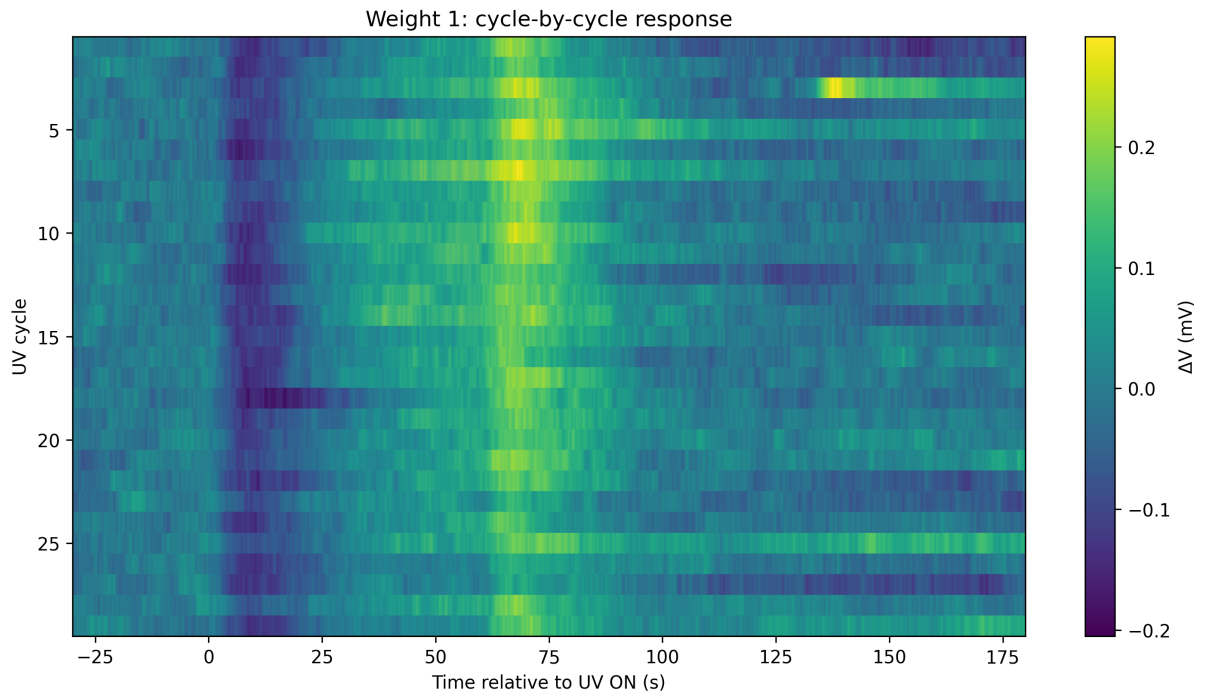


Figure 8: Cycle-by-cycle response matrix for loaded replicate 1.

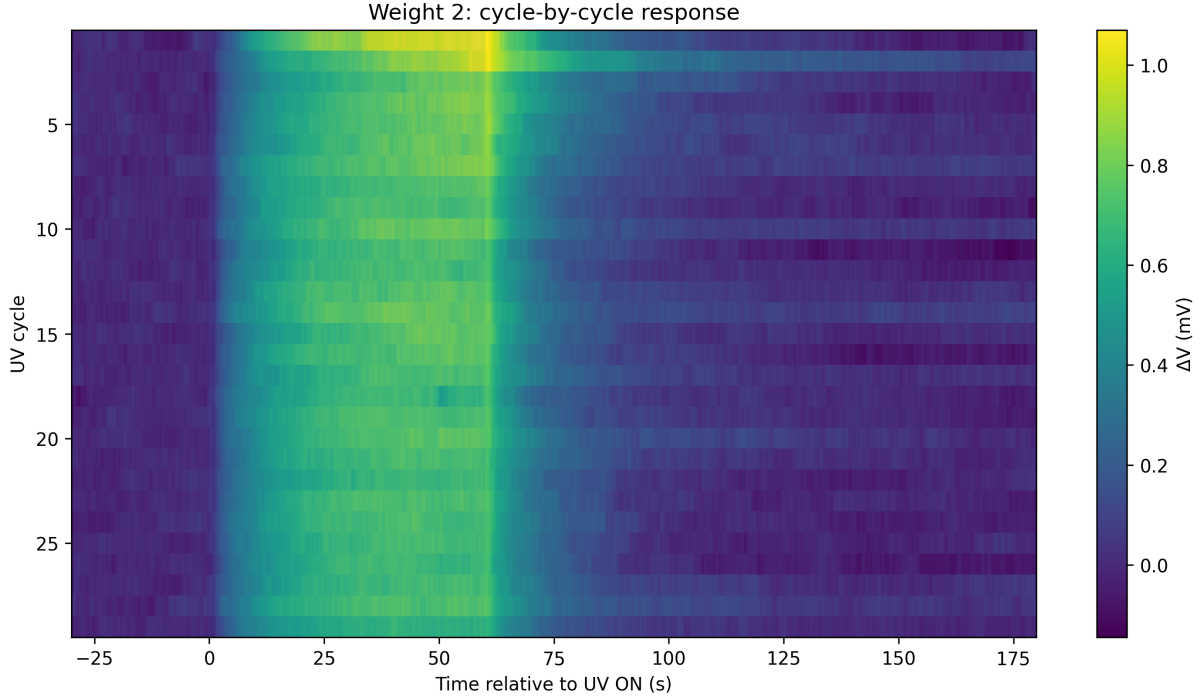


Figure 9: Cycle-by-cycle response matrix for loaded replicate 2.

The heatmaps show that the second loaded recording is particularly repeatable in waveform. Its amplitude decreases over the sequence of repeated exposures, while the temporal structure remains recognizable. The unloaded recording contains several anomalous cycles, notably around cycles 2 and 13–14, which should be retained in the raw data but flagged for quality-control analysis rather than interpreted automatically as large UV responses.

6 Kinetic summary

The numerical summary below is generated directly from the Python analysis script. It is included through a separate generated LaTeX table so that the report remains reproducible when the raw data or processing parameters are changed.

Table 1: Cycle-level kinetic descriptors. Values are means across the available complete-recovery cycles; SD denotes the standard deviation across cycles. These are within-experiment repeated measurements and should not be interpreted as independent biological replicates.

Condition	$A_{\max}$ (mV)	t_{50} (s)	t_{90} (s)	$t_{1/2, \text{rec}}$ (s)	UV AUC (mV s)
No weight	0.471	27.2	45.1	29.9	12.55
Weight 1	0.108	33.9	44.9	29.2	0.38
Weight 2	0.773	7.8	25.1	9.3	37.22

7 Comparison with Mishra et al. (Science Robotics, 2024)

A particularly relevant comparison is provided by Mishra et al., “*Sensorimotor control of robots mediated by electrophysiological measurements of fungal mycelia*,” published in *Science Robotics* in 2024 [2]. The paper demonstrates that fungal mycelia can produce electrophysiological signals in response to environmental stimulation and uses those signals for biohybrid robotic control. In particular, the authors use ultraviolet (UV) stimulation to augment robot gaits and report an experimental electrical response to different UV conditions.

7.1 1. What they actually demonstrate

Their overall chain is approximately

UV light $\rightarrow$ mycelium electrical response $\rightarrow$ signal processing $\rightarrow$ artificial control signal $\rightarrow$ robot movement

The paper demonstrates that UV stimulation produces changes in mycelial electrophysiological activity and that the resulting signals can be exploited for robotic control. Their main emphasis is therefore on the use of fungal electrophysiology as a functional control signal, rather than on a systematic characterization of how mechanical loading modifies the underlying stimulus-response relationship. The paper reports UV tests involving variable UV intensity and exposure time, and quantifies quantities including changes in electrical potential and peak width. For example, the reported measurements include electrical-potential changes of several hundred microvolts and characteristic peak widths on the order of seconds [2].

7.2 2. What we are doing is different

Our experiment is much closer to:

UV stimulation + mechanical loading $\rightarrow$ extracellular electrical response of mycelium.

and we are asking:

How does mechanical loading modify the amplitude and temporal kinetics of the electrical response to UV stimulation?

That is a substantially different scientific question.

	Mishra et al.	Our experiment
Biological material	Mycelium	Mycelium
Electrical measurement	Extracellular electrophysiology	Extracellular voltage
UV stimulation	Yes	Yes
Repeated stimulation	Yes	Yes
Main objective	Biohybrid robot control	Characterize stimulus-response behaviour
UV variable	Intensity / exposure	Repeated standardized 60-s UV pulses
Mechanical load	Not the central variable	Central experimental variable
Agar control	Reference/control experiments	Explicit agar control
Main observables	Spikes, voltage changes, spike timing	Amplitude + kinetics + recovery + AUC
Cycle-by-cycle analysis	Not the main emphasis	Yes
Loading-dependent response	Not investigated as the main question	Yes

Table 2: Conceptual comparison between the Science Robotics study and the present exploratory experiment.

So I would **not** frame our work as simply reproducing Mishra et al.

7.3 3. The most interesting connection: their UV response

This is where our results become particularly interesting.

Mishra et al. show that UV stimulation produces an enhanced electrical response in mycelium and that the response can be exploited as a control signal [2].

Our data independently show a clear UV-locked response. For example, in our **Weight 2** experiment:

- UV ON $\rightarrow$ rapid positive electrical response;
- response reaches roughly 0.77 mV;
- $t_{50} \approx 7.8$ s;
- $t_{90} \approx 25.1$ s;
- recovery half-time ≈ 9.3 s.

Whereas the **no-weight** experiment has:

- $A_{\max} \approx 0.47 \text{ mV}$;
- $t_{50} \approx 27.2 \text{ s}$;
- $t_{90} \approx 45.1 \text{ s}$;
- recovery half-time $\approx 29.9 \text{ s}$.

So the interesting observation is not simply:

“UV causes an electrical signal.”

That is already established by Mishra et al.
It is:

The temporal phenotype of the UV-induced electrical response appears to depend strongly on mechanical loading.

That is much more interesting scientifically.

7.4 4. There is an important difference in timescale

This is something to emphasize if we write a paper.

Mishra et al. are particularly interested in action-potential-like spikes and their use as digital/robotic control signals. Their UV stimulation produces relatively rapid electrical events, with characteristic peak widths on the order of seconds [2].

Our analysis is looking at a different component of the response: the slower transient envelope associated with the 60-s illumination protocol.

Our 60-s illumination produces a **much slower transient envelope**. For example, our no-weight response is approximately:



Whereas their application is much more concerned with detecting the discrete electrical events and converting them into control commands.

This suggests that we may be measuring a **different level of the biological response** rather than competing with their spike-based measurement.

7.5 5. The really interesting result may be Weight 2

Looking at our results in light of their paper changes how we would interpret our experiment.

Initially we were asking:

Does weight increase or decrease the voltage?

I now think that is **too simplistic**.

Our data suggest that loading may alter at least three independent properties:

Amplitude

$A_{\max}$.

Weight 2:

0.773 mV

No weight:

0.471 mV.

Response speed

t_{50} .

Weight 2:

7.8 s

No weight:

27.2 s.

Recovery speed

$t_{1/2,\text{rec}}$.

Weight 2:

9.3 s

No weight:

29.9 s.

So the striking thing is actually:

Weight 2 does not merely produce a larger response; it produces a qualitatively faster response.

That is potentially much more interesting.

7.6 6. And this gives us a better scientific hypothesis

Instead of:

Mechanical loading affects the electrical response of mycelium to UV.

I would formulate the hypothesis as:

Mechanical loading modulates the amplitude and temporal kinetics of UV-induced mycelial electrophysiology

Then we have measurable quantities:

$\{A_{\max}, t_{50}, t_{90}, t_{\text{rec}}, \text{AUC}\}$.

This is much stronger experimentally.

7.7 7. Our agar control becomes especially important

The Science Robotics paper establishes that mycelium responds to UV, but our experiment has another useful element: **agar without mycelium**.

Our agar response does not reproduce the strong sustained positive response seen particularly in Weight 2.

For example, approximately:

	Agar	Weight 2
0–20 s	−0.05 mV	+0.57 mV
40–60 s	−0.09 mV	+1.02 mV
60–80 s	−0.13 mV	+0.64 mV

Table 3: Descriptive comparison of representative baseline-subtracted UV-cycle windows for the agar control and Weight 2 experiment.

So this gives us an important control against simply saying:

$$UV + \text{conductive/moist biological substrate} = \text{voltage artifact}.$$

It supports the interpretation that the large positive response is associated with the **presence of living mycelium**, although it does not by itself establish the precise biological mechanism.

7.8 8. There is also a methodological lesson from their paper

Their work puts considerable emphasis on obtaining stable electrophysiological measurements and shielding the measurement from vibration and electromagnetic interference [2].

That is relevant to us.

Our next experiment should therefore make the distinction between

$$V_{\text{measured}} = V_{\text{biological}} + V_{\text{UV/artifact}} + V_{\text{mechanical}} + V_{\text{electrical noise}}$$

as cleanly as possible.

7.9 9. Where I think our work could go beyond the Science Robotics paper

This is the part I find most promising.

We could construct a **stimulus–mechanics–electrical response map**:

$$(L, I_{\text{UV}}, T_{\text{UV}}) \longrightarrow V(t)$$

where

- L = mechanical load;
- I_{UV} = UV intensity;
- T_{UV} = exposure duration.

and then reduce $V(t)$ to:

$$A, \quad t_{50}, \quad t_{90}, \quad t_{\text{recovery}}, \quad \text{AUC}.$$

That turns the mycelium into something closer to an **engineered living transducer**. Conceptually:

Mechanical state $\rightarrow$ modulation of biological response $\rightarrow$ electrical signal

while Mishra et al. essentially use:

environmental stimulus $\rightarrow$ mycelial electrical signal $\rightarrow$ robotic actuation.

Those are complementary rather than redundant.

7.10 10. One major caveat

We should **not yet claim that loading causes the differences**.

We currently have:

- one no-weight experiment;
- one Weight 1 experiment;
- one Weight 2 experiment;
- repeated UV cycles within each experiment.

The 30 cycles are **not 30 independent biological replicates**.

And, importantly, Weight 1 and Weight 2 behave very differently.

That makes the next experiment particularly important.

I would do something like:

$$0, L_1, L_2, L_3$$

with **at least 3 independent mycelium samples per loading condition**, and perhaps 5–10 UV cycles per sample.

Then we can ask whether

$$A_{\max} = f(L)$$

and

$$t_{50} = g(L)$$

actually emerge as reproducible relationships.

7.11 Overall assessment

The comparison strengthens the rationale for what we are doing.

Mishra et al. have already established the important premise:

mycelium can convert UV stimulation into measurable electrical activity and that activity can be functionally exploited.

Our experiment asks a different and natural next question:

Can the electrical transduction of an environmental stimulus be modulated by the mechanical state of the living fungal material?

Our preliminary data suggest that the answer may be **yes**, particularly because the loading condition appears to affect not only amplitude but also **response and recovery kinetics**.

We would therefore **not change our analysis pipeline**. Instead, the Science Robotics paper can be used to strengthen the scientific framing and make **amplitude + kinetics + mechanical loading** the central story.

8 Combined time-domain and frequency-domain characterization

The time-domain analysis describes amplitude and kinetics of the UV-induced response. We next treat the experiment as a periodic input–output problem because the UV command is deterministic: 60 s ON followed by 180 s OFF, repeated every 240 s. The fundamental stimulus frequency is

$$f_0 = \frac{1}{240} = 0.004\,167\text{ Hz},$$

with harmonics at integer multiples of f_0 .

For each complete UV cycle, the filtered voltage was represented by its complex Fourier coefficient Y_k and the known UV command by U_k . We formed the descriptive harmonic input–output estimate

$$H_k = \frac{Y_k}{U_k}.$$

The magnitude $|H_k|$ describes voltage response per unit UV input at harmonic k , while $\arg(H_k)$ describes the phase relationship. This is a descriptive harmonic response, not yet a conventional linear time-invariant transfer function: the biological system may be nonlinear, time-varying, and adaptive.

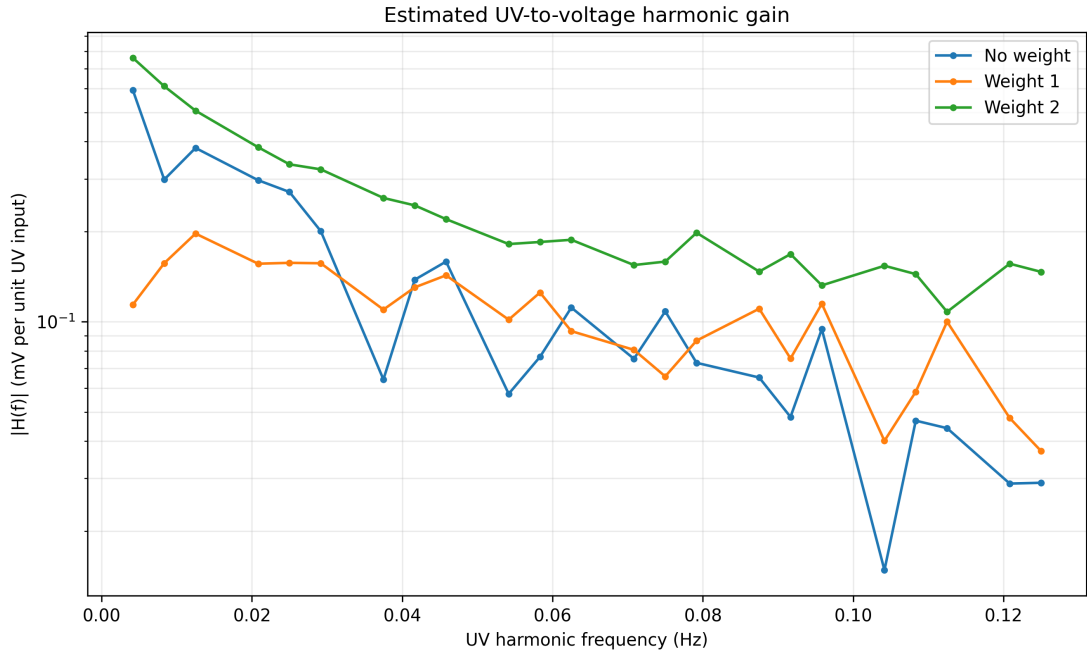


Figure 10: Estimated UV-to-voltage harmonic gain. The Weight 2 recording exhibits substantially larger gain over the first several harmonics.

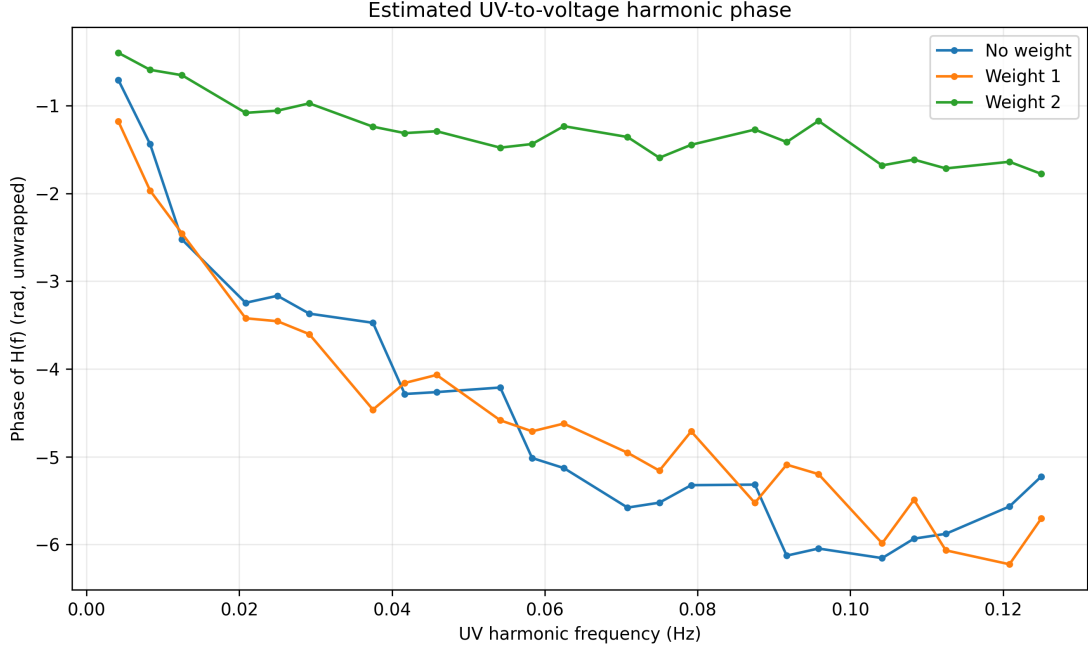


Figure 11: Estimated phase of the UV-to-voltage harmonic response.

8.1 Harmonic gain and coherence

The first-harmonic results provide a compact comparison:

Condition	$ H_1 $ (mV)	Phase (rad)	Coherence
No weight	0.593	-0.706	0.660
Weight 1	0.114	-1.179	0.782
Weight 2	0.760	-0.398	0.977

Table 4: Descriptive first-harmonic UV-to-voltage response. Coherence is magnitude-squared coherence at the UV fundamental.

The Weight 2 experiment is notable because the observations converge across different descriptions of the same response. In the time domain it has the largest response amplitude, shortest t_{50} , shortest recovery half-time, and largest UV AUC. In the frequency domain it also has the largest first-harmonic gain and the highest coherence with the known UV input.

At the first three harmonics, Weight 2 has coherence values of approximately

$$C_1 = 0.977, \quad C_2 = 0.963, \quad C_3 = 0.957.$$

The corresponding first-harmonic coherence for the no-weight experiment is approximately 0.660. The appropriate pilot interpretation is therefore:

The loaded Weight 2 condition exhibited a substantially stronger and more phase-consistent voltage response at the harmonics of the periodic UV stimulus.

This is not evidence of a statistically established loading effect because only one biological recording is available for each condition.

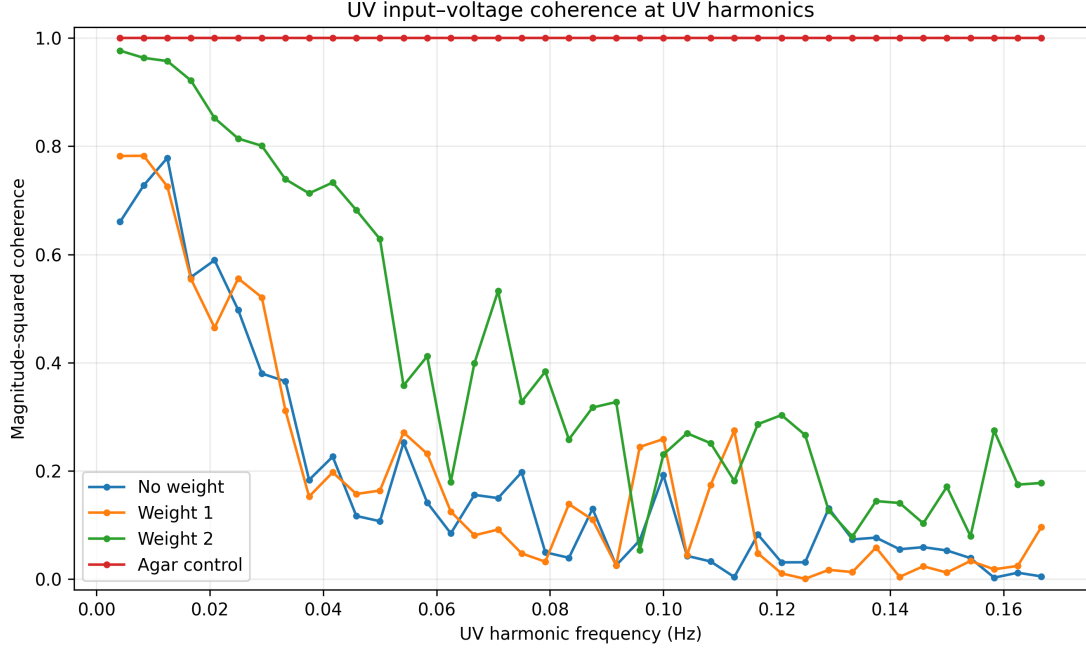


Figure 12: Magnitude-squared coherence between the reconstructed UV input and measured voltage at the UV harmonics.

8.2 Cycle-to-cycle evolution

Harmonic gain was also evaluated separately for each UV cycle. This provides a frequency-domain counterpart to the cycle-by-cycle time-domain metrics and allows possible adaptation or evolution of the response to be identified.

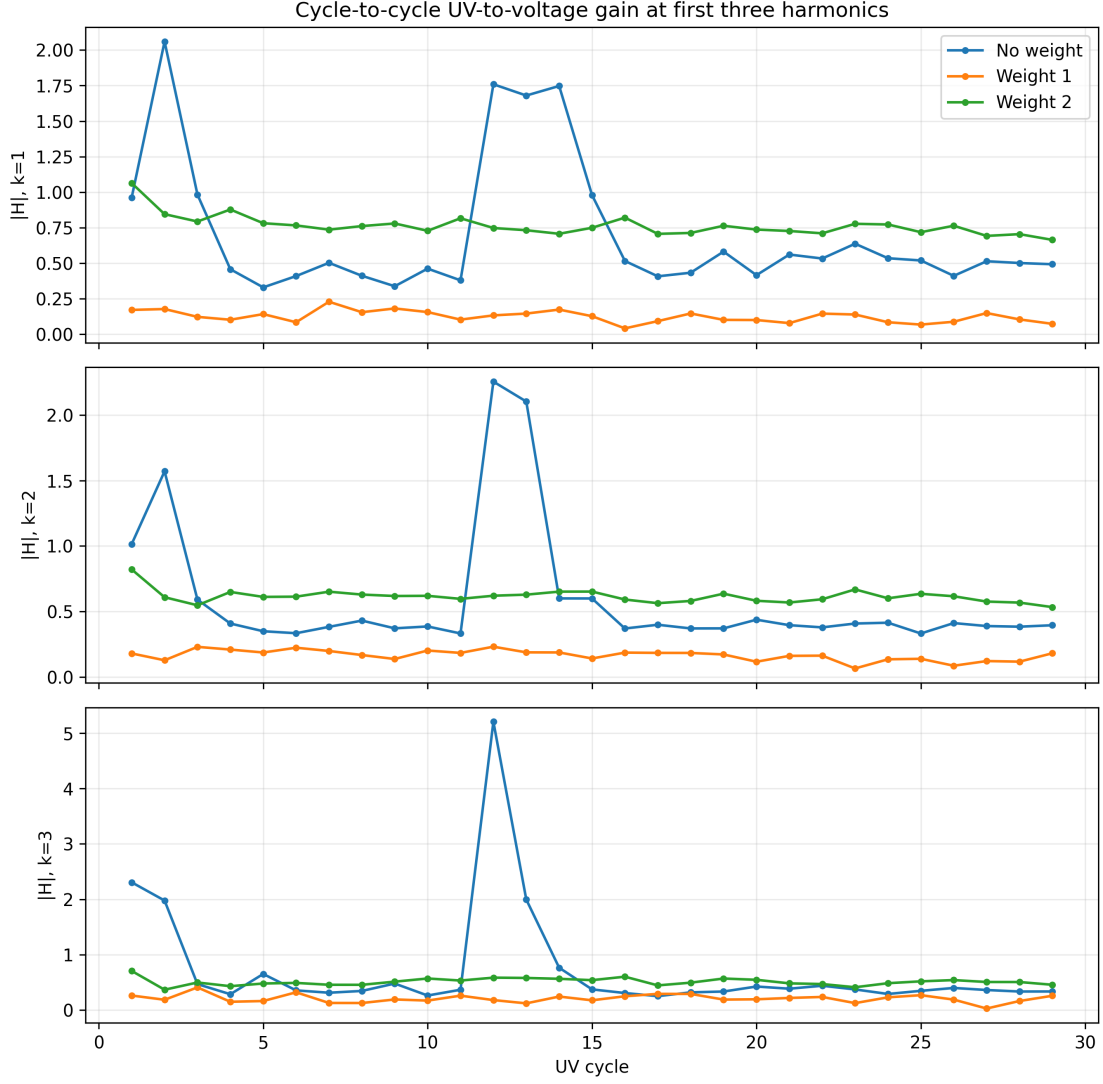


Figure 13: Cycle-to-cycle harmonic gain for the first three UV harmonics.

8.3 Unified stimulus–response description

The two analyses can be viewed as complementary projections of the same experiment:

$$\boxed{\text{UV input} \rightarrow \text{mycelium under mechanical state } L \rightarrow V(t)}.$$

Time-domain observables are

$$\{A_{\max}, t_{50}, t_{90}, t_{\text{rec}}, \text{AUC}\},$$

while frequency-domain observables are

$$\{|H_k|, \arg(H_k), C_k\}.$$

For the current data, Weight 2 simultaneously shows the largest time-domain response amplitude, fastest apparent rise and recovery, largest UV AUC, largest fundamental harmonic gain, high coherence with the periodic UV input, and high phase consistency across repeated cycles. This convergence makes Weight 2 a particularly interesting pilot observation, but it remains a single biological run.

Recent fungal electrophysiology work has likewise used short-time Fourier analysis to extract frequency patterns from extracellular voltage recordings, supporting the methodological

relevance of frequency-domain analysis while using a different experimental system and objective [1].

8.4 Next experimental test

The next experiment should retain the present time-domain measurements while adding harmonic gain and UV–voltage coherence as predefined response descriptors. For each independent sample, several standardized UV cycles should be recorded. The independent experimental unit must remain the biological sample rather than the repeated cycle. With multiple independent samples, we can test whether mechanical loading is associated with reproducible changes in amplitude, kinetics, harmonic gain, and UV–voltage coherence.

The sharpened hypothesis is:

Mechanical state modulates both the temporal kinetics and the frequency-domain coupling of mycelial electri

9 Interpretation and limitations

The current data support the following pilot observations:

1. Repeated UV illumination produces a reproducible, time-dependent extracellular voltage response in the mycelium recordings.
2. The response waveform differs between the three mycelium recordings. In particular, loaded replicate 2 exhibits a strong, predominantly positive and comparatively rapid response.
3. The response amplitude in loaded replicate 2 decreases over the repeated UV sequence, suggesting adaptation of the measured response within that experiment.
4. The agar control does not reproduce the characteristic positive mycelium response, although its short duration limits the strength of this comparison.
5. The two recordings obtained with the same nominal static weight differ substantially. Therefore, the present data cannot by themselves establish a causal effect of static loading.

The 30 UV cycles in a single recording are repeated observations within one experimental run, not 30 independent biological samples. Consequently, formal inference about the effect of static loading should be based on multiple independent mycelium samples per loading condition.

10 Reproducibility

All numerical processing and figure generation are contained in `mycelium_analysis.py`. To reproduce the analysis:

```
python mycelium_analysis.py
```

The script reads the four NumPy files from the `data/` directory and writes figures to `figures/` and numerical results to `results/`.

11 Recommended next experimental step

For a statistically defensible study of static loading, the independent experimental unit should be the mycelium sample rather than the individual UV cycle. A useful design would therefore include several independent samples under zero load and several independent samples under the same static load, with the repeated UV cycles retained as within-sample repeated measurements.

References

- [1] M. Buffi et al., “Detection of electrical signals in fungal mycelia in response to external stimuli,” *iScience*, vol. 28, no. 10, 113484, 2025. doi:10.1016/j.isci.2025.113484
- [2] A. K. Mishra, J. Kim, H. Baghdadi, B. R. Johnson, K. T. Hodge, and R. F. Shepherd, “Sensorimotor control of robots mediated by electrophysiological measurements of fungal mycelia,” *Science Robotics*, vol. 9, no. 93, eadk8019, 2024. doi:10.1126/scirobotics.adk8019